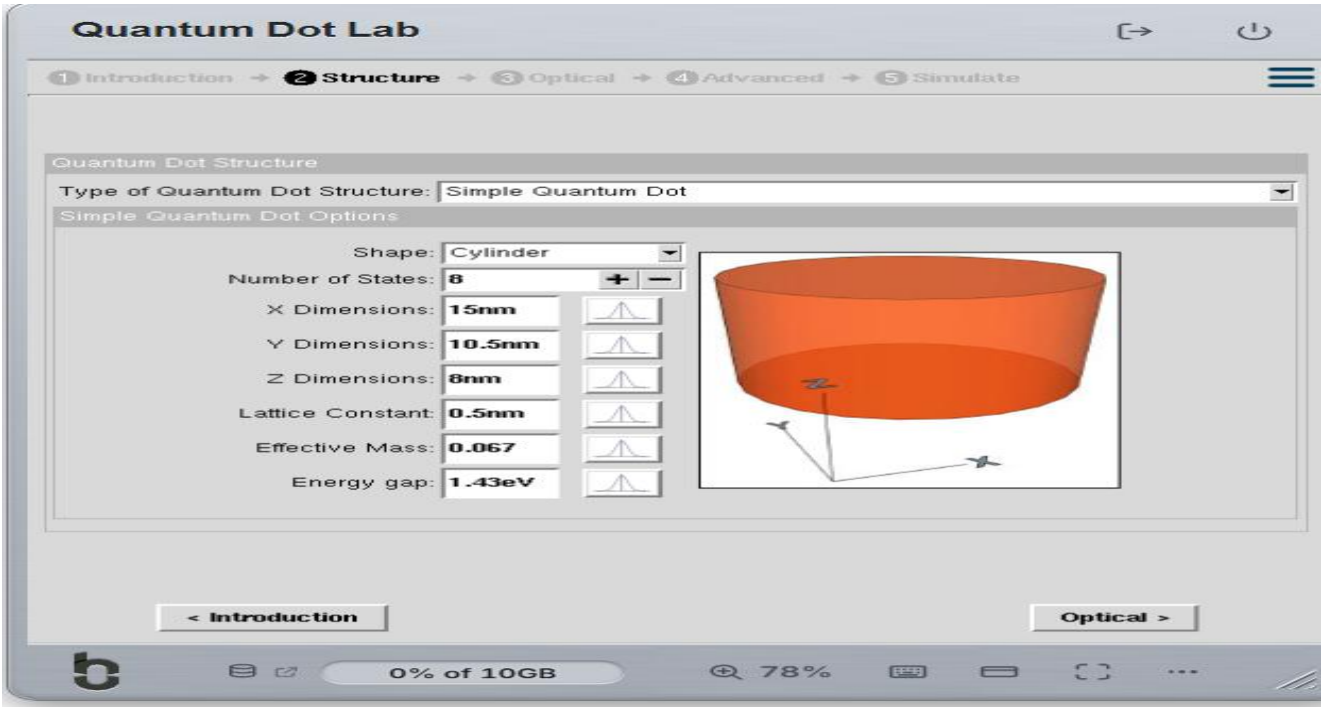
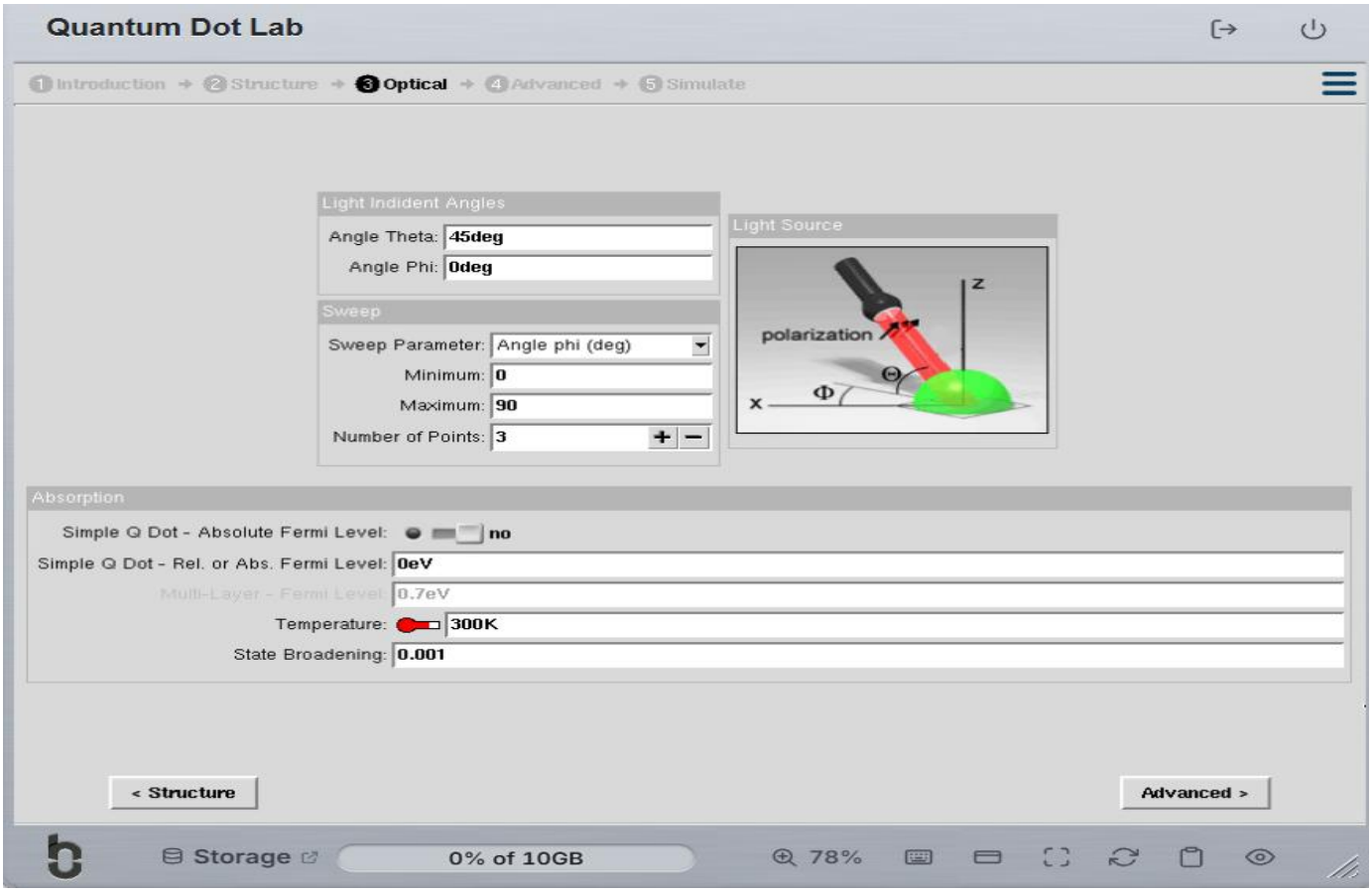
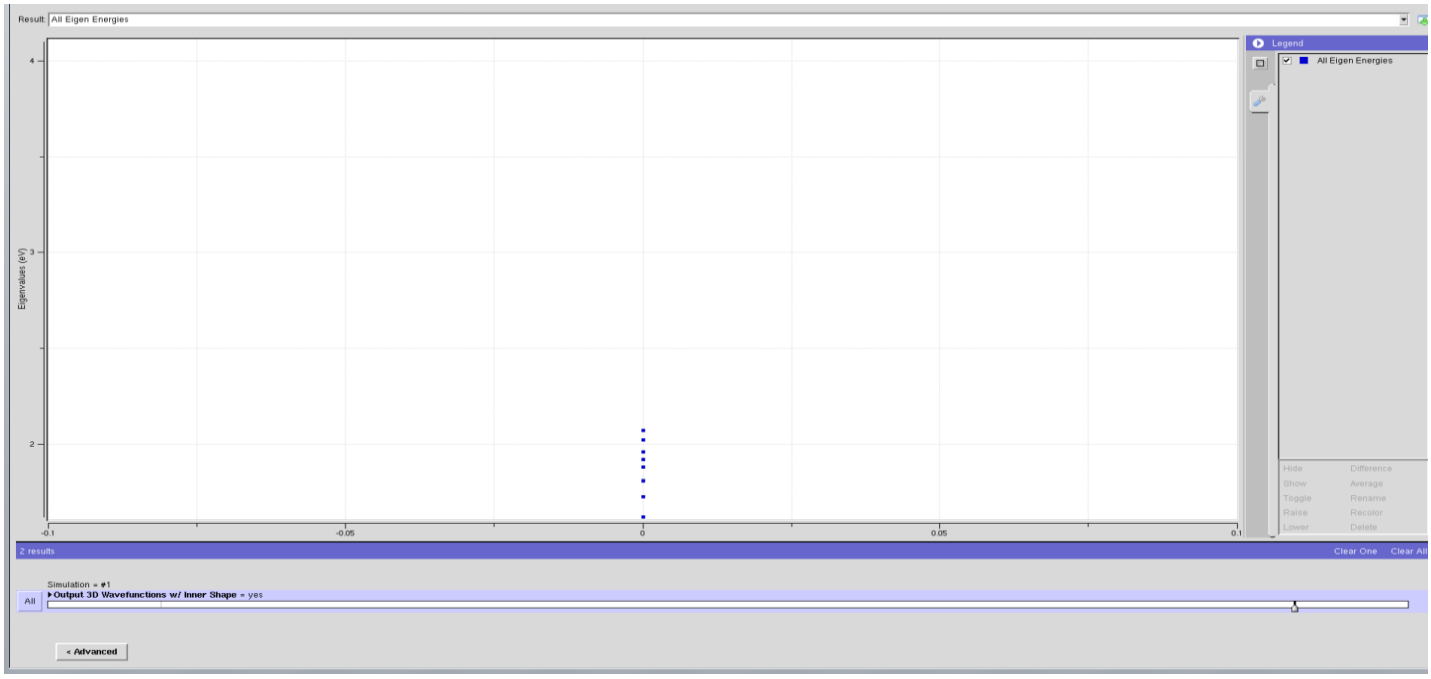
EXP. 3
SIMULATION OF ENERGY STATES OF QUANTUM DOT USING NANOHUB

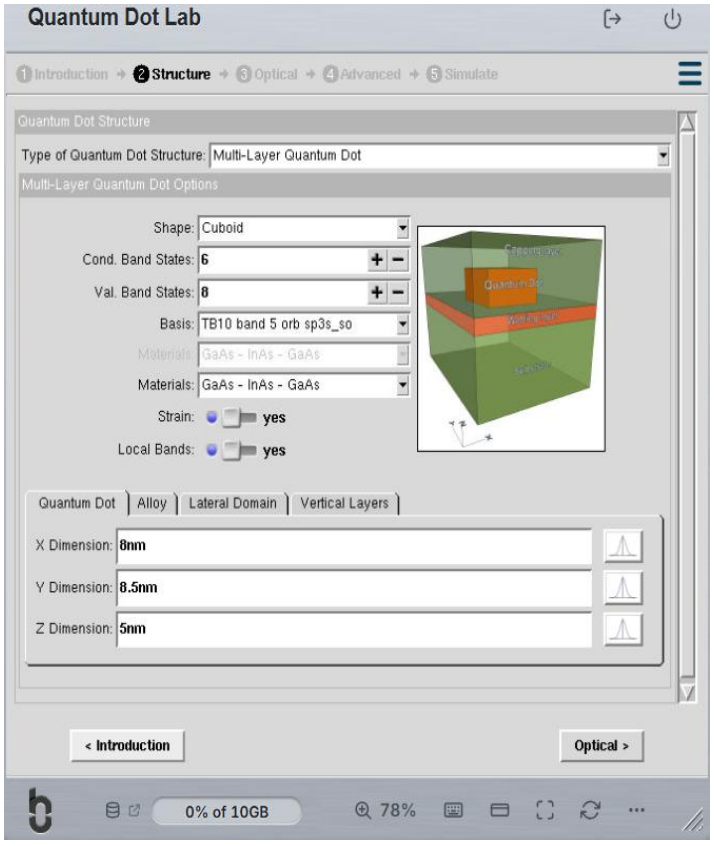
Test case ~ 1

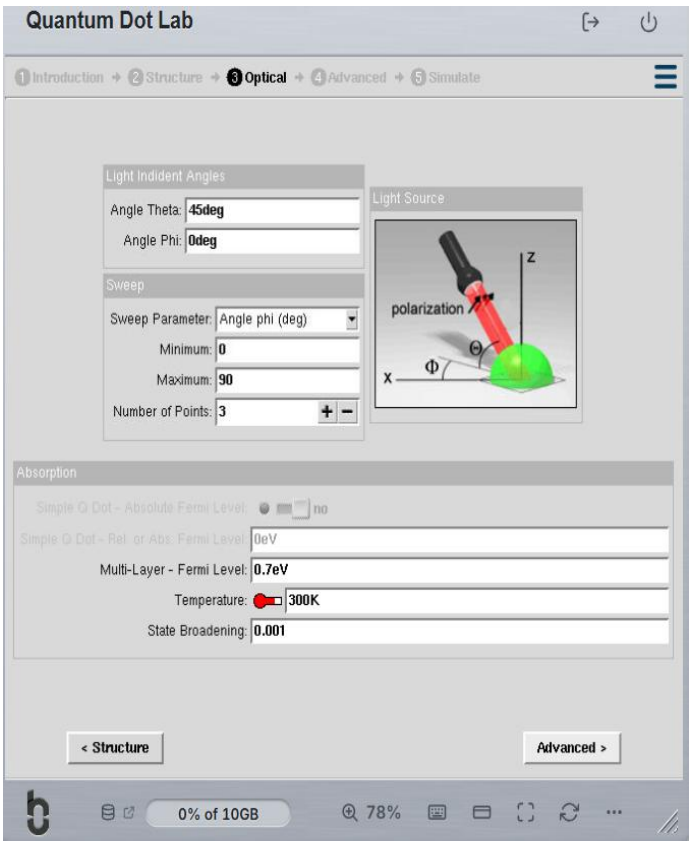


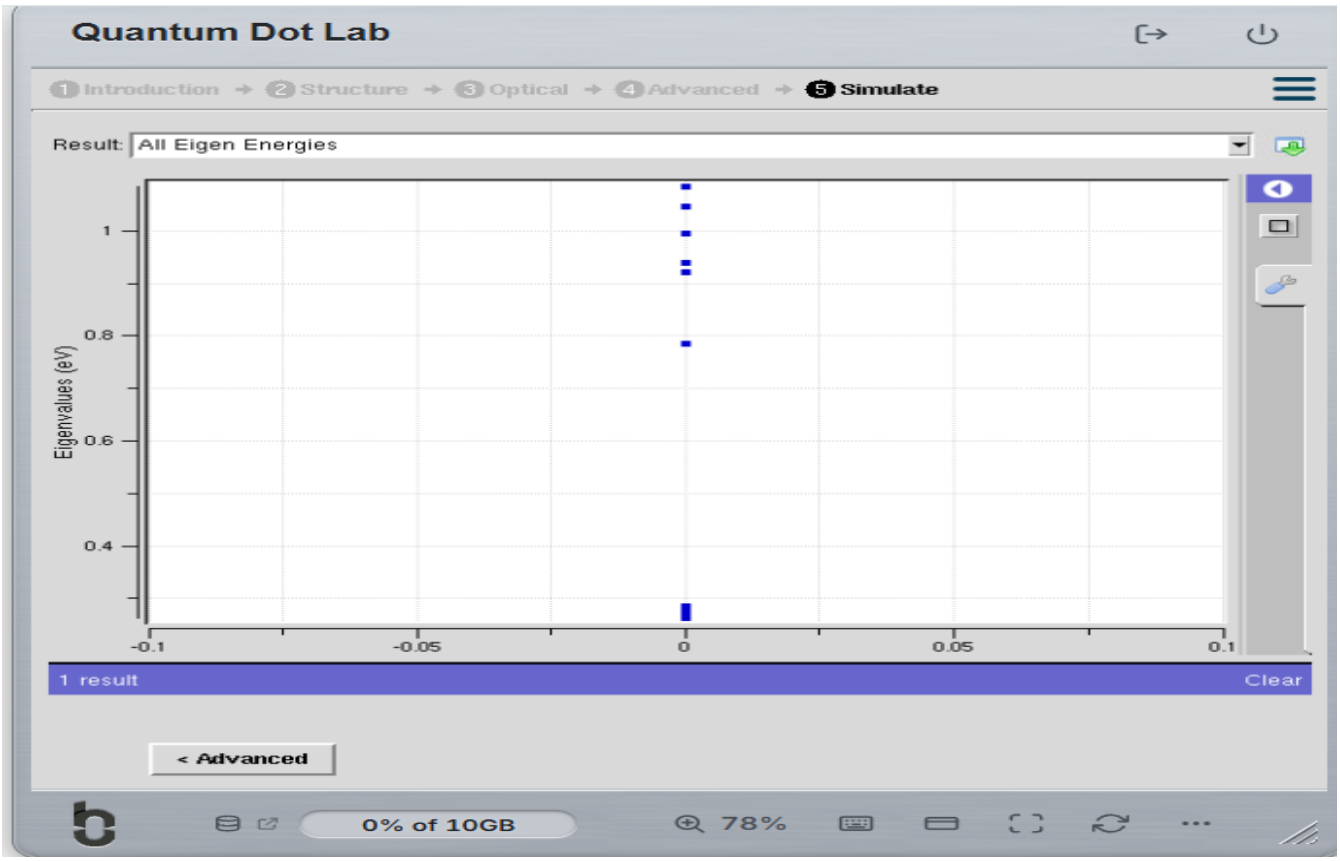



Result: All Eigen Energies
Legend
All Eigen Energies
Eigenvalues (eV)
2 results
Clear One
Clear All
Simulation = #1
Output 3D Wavefunctions w/ Inner Shape = yes
All
< Advanced

Test case ~ 2


Quantum Dot Lab
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Quantum Dot Structure
Type of Quantum Dot Structure: Multi-Layer Quantum Dot
Multi-Layer Quantum Dot Options
Shape: Cuboid
Cond. Band States: 6
Val. Band States: 8
Basis: TB10 band 5 orb sp3s_so
Materials: GaAs - InAs - GaAs
Strain: yes
Local Bands: yes
Quantum Dot
Alloy
Lateral Domain
Vertical Layers
X Dimension: 8nm
Y Dimension: 8.5nm
Z Dimension: 5nm
< Introduction
Optical >
0% of 10GB
78%


Quantum Dot Lab
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Light Incident Angles
Angle Theta: 45deg
Angle Phi: 0deg
Light Source
Sweep
Sweep Parameter: Angle phi (deg)
Minimum: 0
Maximum: 90
Number of Points: 3
polarization
Absorption
Multi-Layer - Fermi Level: 0.7eV
Temperature: 300K
State Broadening: 0.001
< Structure
Advanced >
0% of 10GB
78%


Quantum Dot Lab
1 Introduction → 2 Structure → 3 Optical → 4 Advanced → 5 Simulate
Result: All Eigen Energies
Eigenvalues (eV)
1
0.8
0.6
0.4
-0.1
-0.05
0
0.05
0.1
1 result
Clear
< Advanced
0% of 10GB
78%